A Comparative Study of Statistical and Machine Learning Models for Retail Demand Forecasting

Lilit Babakhanyan
Akian College of Science and Engineering
American University of Armenia
Yerevan, Armenia
lilit_babakhanyan@edu.aua.am

Dalita A. Avanesian
Akian College of Science and Engineering
American University of Armenia
Yerevan, Armenia
davanesyan@aua.am

***Abstract*—Accurate demand forecasting plays a critical role in retail decision-making, enabling businesses to optimize inventory management, staffing, and promotional strategies. This capstone project investigates forecasting methods for predicting weekly retail sales using a large-scale dataset containing multiple stores and departments over several years. The objective is to evaluate and compare baseline models, classical time-series approaches, and machine learning methods in order to identify models capable of capturing trends, seasonality, and holiday-driven demand fluctuations.**

The study begins with comprehensive data preprocessing, including time-series construction, handling missing observations, and feature engineering to incorporate temporal and external variables such as holidays, economic indicators, and store-level characteristics. Exploratory data analysis is conducted to identify long-term trends, seasonal patterns, and demand fluctuations. Baseline forecasting models, including naive, moving average, and seasonal naive approaches, are implemented to establish performance benchmarks.

Building upon these baselines, classical time-series models such as ARIMA and SARIMA are developed to capture temporal dependencies and seasonal structures. The analysis is further extended to machine learning models, including Linear Regression and Random Forest, using engineered features such as lag variables, rolling statistics, and calendar-based attributes.

Results show that SARIMA and Random Forest achieve comparable predictive performance, with SARIMA recording slightly lower percentage error and Random Forest achieving lower absolute error metrics. These findings highlight the importance of modeling seasonality explicitly while demonstrating the potential of machine learning methods in demand forecasting. The study provides actionable insights for selecting forecasting approaches that improve operational efficiency and support data-driven decision-making in retail environments.

***Index Terms*—demand forecasting, retail analytics, time-series forecasting, ARIMA, SARIMA, random forest, machine learning, feature engineering**

I. Introduction

Inventory distortion, the combined cost of overstocks and stockouts, remains a persistent challenge in the retail industry, directly impacting profitability and operational efficiency. According to a report by IHL Group [5], inventory distortion resulted in global losses of approximately $1.73 trillion in 2025, accounting for 6.5% of total retail sales. The magnitude of this loss is comparable to the entire GDP of South Korea. At its core, retail management seeks to achieve two fundamental objectives: profit maximization and cost minimization. While rapid technological advancements have significantly improved forecasting capabilities, challenges still remain. The gap between technological potential and real-world performance highlights the need for more accurate and robust demand forecasting methodologies.

Demand forecasting is the process of estimating future customer demand based on historical data and relevant influencing factors. In retail operations, it plays a central role, as accurate forecasting enables businesses to optimize inventory levels and reduce waste. However, forecasting demand is inherently challenging due to the uncertainty and variability associated with real-world data. The predictability of demand depends on several factors, including how well the contributing factors are understood, the availability and quality of data, the extent to which future patterns resemble the past, and the degree to which forecasting itself influences future outcomes [4]. In retail environments, these challenges are further intensified by external variables such as promotions, seasonality, and economic conditions, all of which can introduce significant and often unpredictable fluctuations in demand.

Historically, the field of demand forecasting was dominated by classical statistical methods, most notably the Autoregressive Integrated Moving Average (ARIMA) framework introduced by Box and Jenkins [1]. These models offer a systematic way to analyze time-series data by identifying patterns in past observations. Extensions such as Seasonal ARIMA (SARIMA) were later developed to account for recurring seasonal cycles common in retail environments, such as holiday peaks and weekly demand patterns. Despite their theoretical elegance and continued relevance as benchmarks, these classical approaches are inherently univariate and rely primarily on historical observations. However, relying solely on historical data limits their ability to capture complex nonlinear relationships and the influence of external factors [3]. Nevertheless, in many real-world applications, including this study, models such as SARIMA remain highly effective when the data exhibits clear trends and seasonality.

The increasing complexity of modern retail, including a large variety of products, frequent promotional activities, and changing economic conditions, has led to a growing use of machine learning approaches. Unlike classical statistical models, machine learning methods, including tree-based approaches

such as Random Forest and gradient-boosted models, are capable of capturing complex nonlinear relationships in demand data and incorporating multiple input variables, making them well-suited for multivariate retail forecasting problems [3].

The effectiveness of these methods has been explored in forecasting competitions. For example, the M3 competition [6] showed that simple statistical models often performed as well as, or better than, more complex machine learning approaches. However, the more recent M5 competition, which focused on Walmart retail data, demonstrated that machine learning models, particularly gradient-boosted trees, could achieve strong performance across many product-store combinations [7].

At the same time, this advantage is not consistent across all cases. The performance of machine learning compared to classical models depends on factors such as the structure of the dataset, the strength of seasonal patterns, and the amount of available data [2], [6]. In particular, Cerqueira et al. [6] show that the relative performance of forecasting methods is strongly influenced by data availability. Their findings suggest that when time-series are short, classical statistical models often outperform machine learning approaches, whereas machine learning methods tend to perform better when larger amounts of data are available.

Based on these considerations, this study evaluates and compares demand forecasting approaches across three methodological families: baseline methods, classical time-series models, and machine learning techniques, applied to a large-scale dataset of weekly Walmart sales across 45 stores and multiple departments. A key focus of the analysis is understanding how data characteristics, particularly the strength of seasonal patterns and the structure of the panel data, influence relative model performance. All models are evaluated using a strict chronological train-test split and consistent evaluation metrics (MAE, RMSE, and MAPE), ensuring comparability of results.

In addition to comparing forecasting accuracy across model families, this study makes several contributions. First, it provides a controlled and consistent comparison of baseline, classical time-series, and machine learning models using a unified experimental framework, ensuring fair evaluation across methods. Second, it empirically demonstrates the impact of data aggregation on forecasting performance, showing how accuracy deteriorates as the level of granularity increases from aggregate to store and department levels. Third, the study highlights how machine learning models, through feature engineering, can implicitly capture temporal patterns such as seasonality, achieving performance comparable to classical models like SARIMA. Finally, it provides practical insight into the limitations of commonly used evaluation metrics such as MAPE in the presence of low or zero sales values, which are prevalent in real-world retail data.

II. Research Objective & Questions

The primary objective of this study is to evaluate and compare three families of demand forecasting models, namely baseline methods, classical statistical time-series models, and machine learning approaches, in a real-world retail context. The analysis uses a large-scale panel dataset of weekly Walmart sales covering 45 stores and multiple departments. The study aims to identify which modeling approach achieves the most accurate and reliable demand forecasts. It also seeks to understand the factors that explain differences in performance. A secondary objective is to examine how methodological choices, especially data aggregation level and selection of evaluation metrics, influence model comparison conclusions.

The study is guided by the following central research question:

Which forecasting approach, classical time-series models or machine learning models, provides more accurate predictions of retail demand in a multi-store, multi-department setting under consistent experimental conditions, and what factors explain the observed differences in performance?

To address this question, the analysis is structured around three sub-questions:

(1) *How do classical time-series models compare to machine learning models in terms of forecasting accuracy, as measured by MAE, RMSE, and MAPE?*

This sub-question establishes the core empirical comparison of the study. All models are evaluated under identical experimental conditions. A strict chronological train-test split ensures temporal consistency. This approach isolates the effect of the modeling technique itself, allowing for a fair comparison across baseline methods, ARIMA, SARIMA, and selected machine learning models such as Random Forest. The results provide insight into how model complexity relates to predictive accuracy in this context.

(2) *To what extent does feature engineering, particularly the inclusion of lag variables and rolling window statistics, enable machine learning models to effectively represent temporal dependencies in retail demand data?*

Unlike classical time-series models, which inherently capture temporal dependencies, machine learning approaches require these patterns to be explicitly encoded through feature construction. This sub-question examines whether engineered features are sufficient for capturing trend and seasonal effects. The findings help clarify when machine learning models can effectively approximate time-series behavior.

(3) *How does model performance differ when evaluated at the aggregated (total sales) level versus the disaggregated (store–department) level, and what are the implications for the interpretation of evaluation metrics?*

Retail sales data often exhibit high variability across individual store–department combinations, including low or intermittent demand. This sub-question explores how evaluation metrics, particularly MAPE, behave under these conditions and how conclusions may differ depending on the level of aggregation. It further examines whether models that perform poorly at the disaggregated level may still produce reliable forecasts when evaluated on aggregated demand, with important implications for real-world decision-making.

III. Dataset Description

A. Dataset Overview

The dataset used in this study is the Walmart Store Sales Forecasting dataset, originally made available through a Kaggle competition hosted by Walmart in 2014. It contains historical weekly sales records for 45 Walmart stores located across different regions of the United States, covering a period of approximately three years from February 2010 to October 2012. Each store is divided into multiple departments, and sales are recorded separately for each store–department combination, resulting in a large-scale panel dataset that captures considerable variation in demand patterns across store types, sizes, and geographic regions.

The dataset is particularly well-suited for this study for several reasons. First, it reflects a real-world retail forecasting problem of meaningful scale and complexity, involving multiple parallel time series rather than a single aggregate series. Second, it includes not only historical sales figures but also a rich set of external variables covering promotional activity, macroeconomic conditions, and regional factors, enabling a multivariate modeling approach. Third, the presence of holiday indicators and promotional markdown fields makes it possible to examine the influence of seasonal and promotional effects on demand, which is a central focus of this study.

B. Data Sources and Structure

The dataset is composed of three separate source files, each capturing a distinct aspect of the retail environment. These files were merged during the preprocessing stage to construct a single unified dataset for modeling.

- `train.csv` contains the core sales records, including the target variable `Weekly_Sales` and a holiday indicator, indexed by store, department, and date.
- `features.csv` contains store- and date-level external variables including temperature, fuel price, promotional markdowns, CPI, and unemployment rate.
- `stores.csv` contains static metadata for each of the 45 stores, including store type and size.

The three files are linked through the `Store` identifier, with `features.csv` and `stores.csv` joined to `train.csv` on the `Store` and `Date` fields to produce the final merged dataset used for analysis.

C. Variables and Features

The variables available across the three source files are described in Tables I, II, and III.

TABLE I
Train Dataset Variables

Column	Description	Type	Role
Store	Store identifier	Int	Grouping
Dept	Department identifier	Int	Grouping
Date	Week of observation	Date	Time index
Weekly_Sales	Weekly revenue (USD)	Float	Target
IsHoliday	Holiday indicator	Bool	Feature

TABLE II
Features Dataset Variables

Column	Description	Type	Role
Store	Store identifier	Int	Grouping
Date	Week of observation	Date	Time index
Temperature	Avg weekly temperature	Float	Regional
Fuel_Price	Fuel price (USD)	Float	Economic
MarkDown1–5	Promo markdowns	Float	Promotional
CPI	Consumer Price Index	Float	Economic
Unemployment	Unemployment rate	Float	Economic
IsHoliday	Holiday indicator	Bool	Feature

TABLE III
Stores Dataset Variables

Column	Description	Type	Role
Store	Store identifier	Int	Grouping
Type	Store type (A, B, C)	String	Categorical
Size	Store size (sq ft)	Int	Feature

It is worth noting that the five markdown variables, `MarkDown1--5`, represent anonymized promotional discount amounts applied during a given week. A missing value indicates that no promotion of that type was active during that period. Since markdown data is only available from late 2011 onward, a substantial portion of earlier observations contain no promotional information. This missingness is addressed during preprocessing, as described in Section IV.

D. Temporal Scope and Frequency

The dataset spans approximately 143 weeks, from February 5, 2010 to October 26, 2012, with observations recorded at a weekly frequency. The training file contains 421,570 records in total, with a mean weekly sales value of approximately $15,981 and considerable variance across stores and departments. Weekly sales values range from a minimum of approximately -$4,989, indicating product returns or accounting adjustments, to a maximum of approximately $693,099, illustrating the substantial heterogeneity in sales volumes across store–department combinations.

The weekly granularity of the data is well-suited for capturing seasonal patterns such as holiday spikes and monthly demand cycles, while also providing sufficient historical depth to support the construction of lag features and rolling window statistics used in the machine learning models.

E. Panel Structure and Aggregation Levels

A defining characteristic of this dataset is its panel structure. Rather than representing a single time series, the data comprises multiple parallel time series corresponding to individual store–department combinations. Each combination constitutes an independent demand signal with its own sales trajectory, seasonal pattern, and response to external variables. For example, Department 1 in Store 1 has a distinct sales history from Department 1 in Store 2, even though both are recorded over the same time period.

This structure introduces an important methodological distinction that is central to this study: the difference between

disaggregated and aggregated evaluation. At the disaggregated level, models are trained and evaluated on individual store–department series, preserving granularity but exposing the analysis to high variability and, in many cases, near-zero or intermittent sales values. At the aggregated level, forecasts are summed across all store–department combinations to produce a single total sales series, which tends to be smoother and more stable. Both levels of evaluation are employed in this study, and the implications of this choice for metric interpretation are examined in the results and discussion sections.

IV. Data Preprocessing

Prior to modeling, the raw dataset underwent a series of preprocessing steps to produce a clean, structured, and temporally consistent dataset suitable for both classical time-series and machine learning models.

A. Data Merging

The three source files described in Section III were merged into a single unified dataset. The `features` file was joined to the `train` file using `Store` and `Date` as common keys, and the `stores` file was subsequently joined on the `Store` identifier. A left join was used in both cases to preserve all sales records.

The resulting merged dataset contains 421,570 observations and 17 columns, combining sales figures, external features, and store metadata into a single analytical table.

B. Date Conversion and Sorting

The `Date` column was converted from string format to a datetime object to enable chronological operations and time-based indexing. The dataset was then sorted by `Store`, `Dept`, and `Date` to ensure that each store–department series is ordered chronologically.

This ordering is essential for the correct computation of lag and rolling features, as well as for the chronological train–test split described in Section IV-E.

C. Handling Missing Values

The primary source of missing data in the merged dataset was the five promotional markdown variables (`MarkDown1--5`), which account for a substantial proportion of missing entries. As discussed in Section III, markdown data is only available from late 2011 onward, meaning that earlier observations contain no promotional information by design rather than due to data quality issues.

Missing markdown values were therefore filled with zero, reflecting the assumption that no promotional discount was applied during those weeks. This is a standard and well-justified treatment in retail forecasting, where the absence of a promotion is itself informative [3].

In the merged dataset used for modeling, no other variables contained missing values.

D. Feature Engineering

To enable machine learning models to capture temporal patterns in demand, a set of time-based and lagged features was constructed for each store–department series. This step is necessary because machine learning models do not inherently account for temporal dependencies and must instead rely on engineered features.

The following features were created:

- Calendar features: Year, Month, Week of Year, and Quarter extracted from the `Date` column
- Lag features: Weekly sales lagged by 1, 2, 4, 12, and 52 weeks
- Rolling features: 4-week and 12-week rolling means, along with a 4-week rolling standard deviation

All features were computed strictly within each store–department group to preserve the independence of individual time series. Rows with missing values introduced by lag and rolling operations (i.e., early observations with insufficient history) were removed.

As a result, the final dataset used for machine learning models contains approximately 261,000 observations. These engineered features were used exclusively for machine learning models. Classical time-series models were applied to aggregated data without feature augmentation, relying on their inherent ability to model temporal structure.

E. Train–Test Split and Data Leakage Prevention

The dataset was partitioned into training and testing sets using a strictly chronological split. Observations up to and including April 10, 2012 were assigned to the training set, and all subsequent observations were reserved for testing. This resulted in approximately 176,000 training observations and 84,000 test observations.

This approach is essential in time-series forecasting to simulate realistic prediction scenarios and prevent data leakage. Unlike random splits commonly used in cross-sectional settings, a chronological split ensures that future information is not used during model training.

The same temporal boundary was applied consistently across all models to ensure fair and directly comparable performance evaluation.

V. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify key patterns, trends, and structural characteristics in the dataset that inform both modeling choices and the interpretation of results. The analysis focuses on four aspects: aggregate demand behavior, seasonality and holiday effects, store-level variation, and the role of external variables.

A. Aggregate Sales Trend

Figure 1 presents total weekly sales aggregated across all stores and departments. The series exhibits a relatively stable baseline with clear cyclical fluctuations and no strong long-term trend.

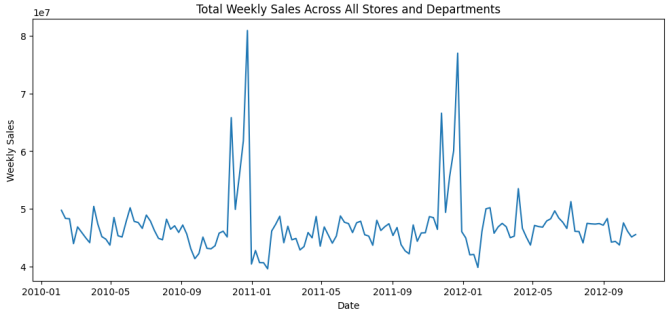


Fig. 1. Total weekly sales aggregated across all stores and departments

The most prominent feature is the presence of sharp, recurring spikes at the end of each year, corresponding to major holiday periods such as Thanksgiving and Christmas. These spikes are consistent across all years and represent the largest deviations from baseline demand.

This stability, combined with strong recurring peaks, suggests that aggregate demand is highly seasonal and relatively predictable. It also supports the use of models capable of capturing annual seasonality, such as SARIMA, and motivates the inclusion of a 52-week lag in machine learning models.

B. Seasonal Patterns and Holiday Effects

Monthly aggregation of sales (Figure 2) confirms a clear seasonal pattern. Demand remains relatively stable throughout most of the year, followed by a significant increase in November and December, with December exhibiting the highest average sales. A noticeable decline occurs in January, reflecting a post-holiday drop in demand.

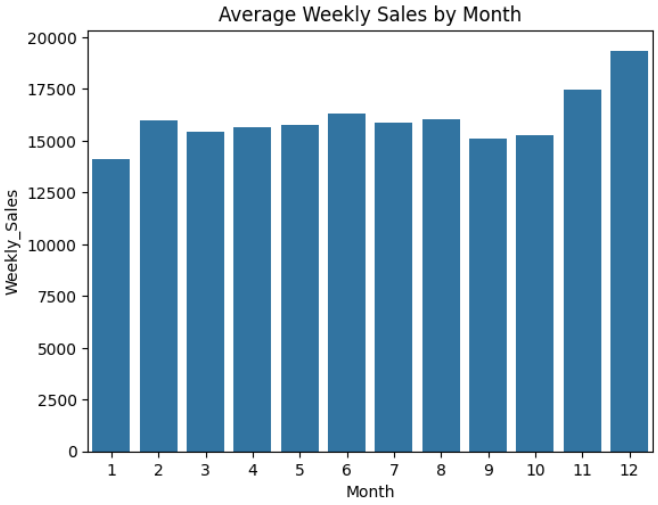


Fig. 2. Bar plot of average weekly sales by month

Holiday effects were further examined by comparing sales during holiday and non-holiday weeks. While average sales are higher during holiday periods, the increase is not uniform across observations and is primarily driven by a small number of extreme values.

This indicates that holiday effects are concentrated around specific events rather than representing a consistent uplift. As a result, a simple binary holiday indicator captures only part of the underlying variation in demand.

C. Store-Level Heterogeneity

Analysis of individual store time series (Figures 3 and 4) reveals substantial variation in sales levels across stores. Larger stores generate significantly higher revenue, while smaller stores operate at a much lower scale.

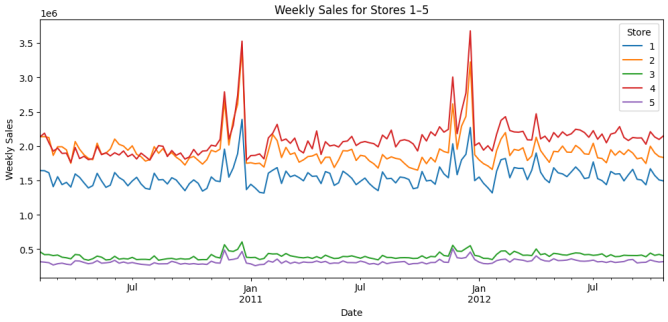


Fig. 3. Weekly sales for Stores 1–5

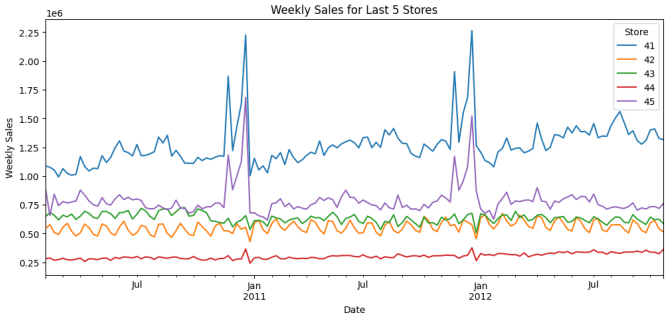


Fig. 4. Weekly sales for Stores 41–45

Despite these differences, the overall temporal patterns, including year-end spikes and mid-year stability, are consistent across stores. This suggests that while demand magnitude varies, the underlying seasonal structure is shared.

This finding supports the use of panel-based modeling approaches and indicates that features capturing seasonality and recent demand trends are likely to generalize well across stores.

D. Relationship Between External Variables and Sales

A correlation analysis between macroeconomic variables (Temperature, Fuel_Price, CPI, and Unemployment) and weekly sales reveals near-zero linear relationships across all variables (Figure 5).

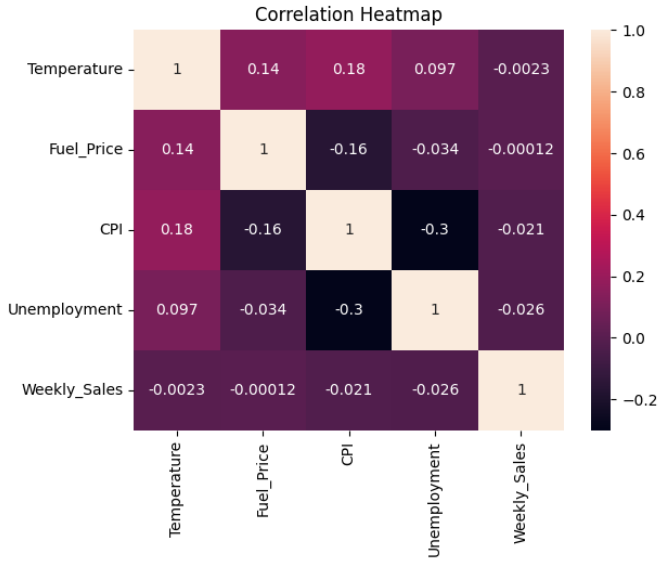


Fig. 5. Correlation matrix between external variables and weekly sales

This suggests that these variables do not have a strong direct linear effect on demand. However, this does not imply that they are uninformative. Their influence may be nonlinear or dependent on interactions with other variables.

This observation suggests that adding these external variables may not necessarily improve forecasting through simple linear relationships. However, tree-based models such as Random Forest may still capture nonlinear or interaction effects.

E. Summary of Key Findings

The exploratory analysis reveals several key insights that inform the modeling strategy of this study. First, aggregate weekly sales are relatively stable and exhibit strong annual seasonality, primarily driven by year-end holiday periods. Second, holiday effects are present but uneven, as increases in demand are driven by a limited number of extreme events rather than a consistent uplift across all observations. Third, there is substantial variation in sales levels across stores, while seasonal patterns are broadly similar but not uniform. Finally, external variables show little evidence of strong linear relationships with weekly sales, suggesting that their influence is likely nonlinear or context-dependent.

These observations collectively motivate the choice of models and evaluation strategy adopted in the following sections.

VI. Methodology

This section describes the forecasting models evaluated in this study, organized into three families: baseline methods, classical time-series models, and machine learning models.

All models are evaluated using a shared chronological test period and common evaluation metrics, while differing in the level of aggregation appropriate to each modeling approach. The section concludes with a description of the evaluation metrics used to assess model performance.

A. Baseline Models

Baseline models serve as reference points against which more complex approaches are evaluated. In forecasting, a more sophisticated model is only justified if it provides a meaningful improvement in accuracy over simpler alternatives [4]. Three baseline methods are considered.

Naive Model.

The naive forecasting method assumes that the best estimate of future demand is the most recently observed value:

$$\hat{y}_{T+h} = y_T \tag{1}$$

Despite its simplicity, the naive model provides a useful lower bound for forecasting performance.

Moving Average Model.

The moving average model forecasts demand as the average of a fixed number of recent observations. A four-week moving average is used:

$$\hat{y}_{T+1} = \frac{1}{4}\sum_{i=0}^{3} y_{T-i} \tag{2}$$

This approach smooths short-term fluctuations and provides a stable estimate of recent demand.

Seasonal Naive Model.

The seasonal naive model uses the value from the same period in the previous seasonal cycle:

$$\hat{y}_{T+h} = y_{T+h-52} \tag{3}$$

This method directly captures annual seasonality and serves as a strong benchmark in datasets with recurring patterns, such as retail demand.

B. Classical Time-Series Models

Classical time-series models are applied to the aggregated total sales series, where demand exhibits strong and stable seasonal patterns.

ARIMA.

The Autoregressive Integrated Moving Average (ARIMA) model provides a framework for modeling temporal dependence using past values and past errors. Model parameters (p, d, q) were selected by minimizing the Akaike Information Criterion (AIC), resulting in an ARIMA(5,0,0) specification.

While ARIMA captures short-term dependencies effectively, it does not explicitly account for seasonal patterns, which limits its performance in strongly seasonal datasets.

SARIMA.

Seasonal ARIMA (SARIMA) extends ARIMA by incorporating seasonal components to model recurring patterns at a fixed frequency. In this study, the model was specified as SARIMA$(2,0,2)(1,0,0)_{52}$, where the first set of parameters represents the non-seasonal components and the second set captures the seasonal structure.

This directly corresponds to the strong annual seasonality observed in Section V. By modeling both short-term dynamics and seasonal structure, SARIMA is well-suited to capturing the year-end demand spikes present in the data.

C. Machine Learning Models

Unlike classical time-series models, machine learning models do not inherently capture temporal dependencies. Instead, temporal structure is introduced through the engineered lag and rolling features described in Section IV-D.

Although evaluation is performed at the aggregated level for consistency with time-series models, machine learning models are trained on disaggregated store–department data and then aggregated. This allows them to leverage cross-sectional variation and richer feature information. However, this also introduces a structural difference in how models are trained and evaluated compared to classical time-series approaches, which should be considered when interpreting comparative results.

Linear Regression.

Linear regression models the relationship between the target variable and input features as a linear combination:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n \tag{4}$$

While simple and interpretable, this model assumes linear relationships between inputs and the target, which can be restrictive in retail demand forecasting.

Random Forest.

Random Forest is an ensemble method that constructs multiple decision trees and averages their predictions:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x) \tag{5}$$

This approach reduces variance and improves robustness by combining predictions from many trees trained on bootstrapped samples of the data.

Random Forest can capture nonlinear relationships and interactions between features without requiring explicit specification, making it well-suited for complex retail demand patterns. The Random Forest model was implemented using 100 decision trees, with hyperparameters configured to control model complexity and prevent overfitting. Specifically, the model used a maximum tree depth of 10, a minimum of 10 samples required to split an internal node, and a minimum of 5 samples required at each leaf node. All other parameters were kept at their default values.

D. Evaluation Metrics

Model performance is evaluated using three complementary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Mean Absolute Error (MAE).

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^{n} |y_t - \hat{y}_t| \tag{6}$$

MAE measures the average magnitude of forecast errors in the original unit of the data and is directly interpretable.

Root Mean Squared Error (RMSE).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^{n} (y_t - \hat{y}_t)^2} \tag{7}$$

RMSE penalizes large errors more heavily than MAE, making it more sensitive to outliers.

Mean Absolute Percentage Error (MAPE).

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^{n} \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \tag{8}$$

However, MAPE has a known limitation: when actual values are zero or near zero, it becomes unstable or undefined. This is particularly relevant in retail datasets with low-volume series.

To address this, MAPE is computed only for observations where $y_t > 0$. Additionally, it is primarily interpreted at the aggregated level, where the metric behaves more reliably.

VII. Experimental Setup

This section describes the experimental design used to evaluate the forecasting models introduced in Section VI. It outlines the data split strategy, modeling approach for each model family, aggregation procedure, and implementation environment.

A. Train–Test Split

All models were evaluated using a single chronological train–test split applied consistently across model families. As described in Section IV-E, the split date was set to April 10, 2012, corresponding to an approximate 80/20 partition of the original dataset. This approach prioritizes comparability across models while still preserving realistic chronological ordering. Observations up to and including the split date were used for training, while all subsequent observations were reserved for testing.

Because machine learning models use lagged and rolling features, the effective split proportion differs slightly after preprocessing. However, the same chronological boundary was preserved across all model families to ensure consistency and prevent data leakage.

This approach reflects realistic forecasting conditions and prevents data leakage, ensuring that future observations do not influence model training. The use of a common split boundary across all models ensures a fair and consistent comparison.

B. Baseline and Classical Time-Series Models

Baseline and classical time-series models were applied to a single aggregated time series representing total weekly sales across all stores and departments. This aggregation produced a univariate series of 143 weekly observations.

This design choice was motivated by both practical and methodological considerations. First, ARIMA-type models are univariate and would require fitting thousands of separate models at the store–department level, which is beyond the scope of this study. Second, the aggregated series is smoother and exhibits clearer seasonal patterns, making it more suitable for classical time-series modeling.

Model selection for ARIMA and SARIMA was performed using the `auto_arima` function from the `pmdarima` library, which selects parameters by minimizing the Akaike Information Criterion (AIC). The best-performing model was ARIMA(5,0,0), while SARIMA was estimated with a seasonal period of $s = 52$ to capture annual seasonality. All classical models were implemented using the `statsmodels` library.

C. Machine Learning Models

Machine learning models were trained on the disaggregated store–department dataset, consisting of approximately 261,000 observations and multiple engineered features described in Section IV-D.

Categorical variables, specifically store type, were encoded using one-hot encoding. Continuous variables were standardized using a standard scaler. These preprocessing steps were implemented within a `scikit-learn` Pipeline to ensure that transformations were fitted only on the training data and applied consistently to the test data, preventing data leakage.

The Linear Regression model was implemented using `scikit-learn` with default parameters. The Random Forest model was implemented using `RandomForestRegressor` with 100 estimators, a maximum depth of 10, a minimum of 10 samples required to split a node, and a minimum of 5 samples per leaf. These parameters were selected manually to balance model complexity and generalization. A fixed random state of 42 was used to ensure reproducibility.

D. Aggregation of Machine Learning Predictions

To enable direct comparison with classical models, which operate on aggregated data, predictions from machine learning models were summed across all store–department combinations to produce total weekly sales forecasts:

$$\hat{Y}_T = \sum_{s,d} \hat{y}_{s,d,T} \tag{9}$$

The same aggregation was applied to the actual test values to produce the ground truth series.

This approach allows machine learning models to be evaluated at both the disaggregated and aggregated levels and enables analysis of how aggregation influences apparent model performance, which is central to the third research question.

It is important to note that this experimental design introduces a degree of asymmetry in the comparison between model families. Classical time-series models, including SARIMA, are trained directly on aggregated total sales data, while machine learning models are trained on disaggregated store–department data and subsequently aggregated for evaluation. Although this reflects common practical applications of these methods, it means that the models operate on different data structures and levels of information. This difference should be considered when interpreting comparative performance results.

E. Evaluation Procedure

All models were evaluated using MAE, RMSE, and MAPE, as defined in Section VI-D.

For baseline and classical models, metrics were computed directly on the aggregated series. For machine learning models, metrics were computed at both the disaggregated and aggregated levels.

MAPE was computed only for observations with strictly positive sales values. At the disaggregated level, the presence of near-zero values leads to unstable and inflated MAPE values. Therefore, MAPE is primarily interpreted at the aggregated level, where it provides more reliable comparisons across models. MAE and RMSE are reported at the aggregated level for all models to ensure consistency.

F. Software and Reproducibility

All experiments were conducted in Python 3 using Google Colab, with data stored in Google Drive. The primary libraries used include:

- `pandas` and `NumPy` for data manipulation
- `statsmodels` for ARIMA and SARIMA modeling
- `pmdarima` for automated parameter selection
- `scikit-learn` for machine learning models, preprocessing pipelines, and evaluation metrics
- `matplotlib` for visualization

A fixed random state of 42 was used for all stochastic components to ensure reproducibility of results.

VIII. Results

This section presents the forecasting results for all models evaluated in this study. Results are reported on the held-out test set covering the period from April 2012 to October 2012.

For baseline and classical time-series models, metrics are computed directly on the aggregated total weekly sales series. For machine learning models, performance is evaluated at both the disaggregated store–department level and the aggregated level. Aggregated results are used as the primary basis for cross-model comparison.

A. Baseline Results

Table IV presents the performance of the three baseline models on the aggregated total sales series.

TABLE IV
Baseline Model Performance (Aggregated)

Model	MAE	RMSE	MAPE
Naive	1,648,526	2,419,673	3.52%
Moving Average (4-week)	1,591,870	1,901,638	3.44%
Seasonal Naive	1,247,205	1,562,916	2.65%

The Seasonal Naive model achieves the strongest performance among the baselines across all metrics. In contrast, the standard Naive model performs worst, reflecting its inability to capture temporal structure. The Moving Average model provides a modest improvement but remains substantially weaker than the Seasonal Naive.

This result highlights the dominant role of annual seasonality in the dataset. Simply repeating demand from the same week in the previous year provides a strong predictive signal and establishes a meaningful performance benchmark.

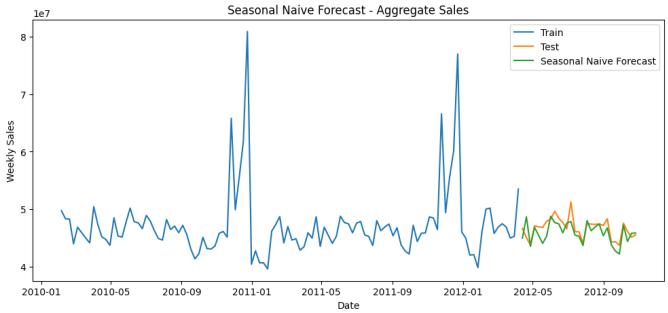


Fig. 6. Seasonal Naive forecast compared to actual values

Figure 6 illustrates the Seasonal Naive forecast compared to actual values. The model captures the overall level and seasonal movement reasonably well but lacks the flexibility to adjust to local fluctuations.

B. Classical Time-Series Results

Table V presents the performance of the classical time-series models.

TABLE V
CLASSICAL TIME-SERIES MODEL PERFORMANCE (AGGREGATED)

Model	MAE	RMSE	MAPE
ARIMA (5,0,0)	1,673,011	2,136,956	3.62%
SARIMA	1,034,871	1,443,201	2.19%

The SARIMA model achieves the lowest MAPE among all models and performs comparably to Random Forest in terms of MAE and RMSE, indicating similar overall predictive accuracy.

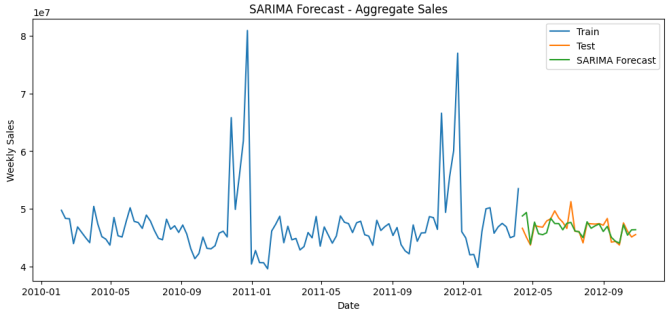


Fig. 7. SARIMA forecast compared to actual sales

Figure 7 shows the SARIMA forecast against actual sales. The model closely tracks both the baseline demand and seasonal variations, demonstrating its ability to capture the dominant annual cycle identified in the EDA.

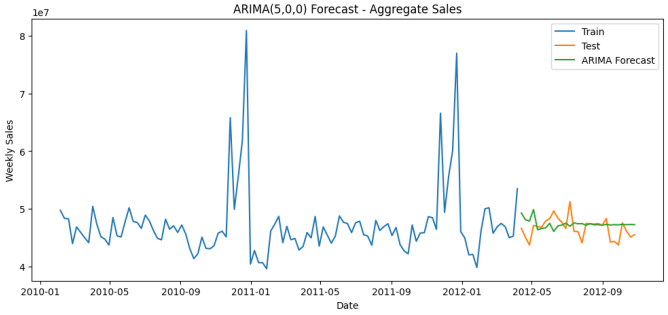


Fig. 8. ARIMA forecast compared to actual sales

By contrast, ARIMA (5,0,0) performs worse than all baseline models. As shown in Figure 8, its forecasts remain relatively flat and fail to capture seasonal peaks, confirming that non-seasonal models are insufficient in this setting.

The contrast between ARIMA and SARIMA isolates seasonality as the primary driver of forecasting accuracy, providing strong empirical support for the importance of seasonal modeling.

C. Machine Learning Results

1) Disaggregated Results: Table VI presents the performance of the machine learning models at the store–department level.

TABLE VI
MACHINE LEARNING MODEL PERFORMANCE (DISAGGREGATED)

Model	MAE	RMSE	MAPE
Linear Regression	1,764	3,198	3,442.54%
Random Forest	1,329	2,861	245.26%

Random Forest outperforms Linear Regression on both MAE and RMSE. However, MAPE values at this level are extremely inflated and not meaningful for comparison.

This instability arises from near-zero sales values, where small absolute errors produce extremely large percentage errors. As a result, MAPE is not reliable at the disaggregated level, confirming that evaluation metrics must be interpreted in the context of data characteristics.

2) Aggregated Results: Table VII presents the performance of machine learning models after aggregating predictions.

TABLE VII
MACHINE LEARNING MODEL PERFORMANCE (AGGREGATED)

Model	MAE	RMSE	MAPE
Linear Regression	2,263,158	2,576,671	4.91%
Random Forest	1,028,661	1,352,851	2.22%

At the aggregated level, results are significantly more stable. Random Forest achieves strong performance, ranking close to SARIMA overall and achieving the lowest MAE and RMSE among all models.

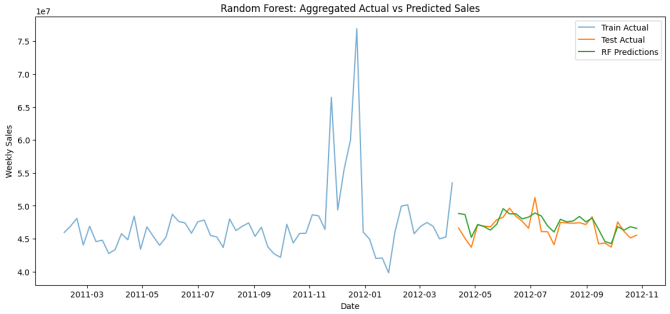


Fig. 9. Random Forest predictions compared to actual values

Figure 9 shows the Random Forest predictions against actual values. The model captures overall demand patterns effectively, though with slightly less precision in seasonal peaks compared to SARIMA.

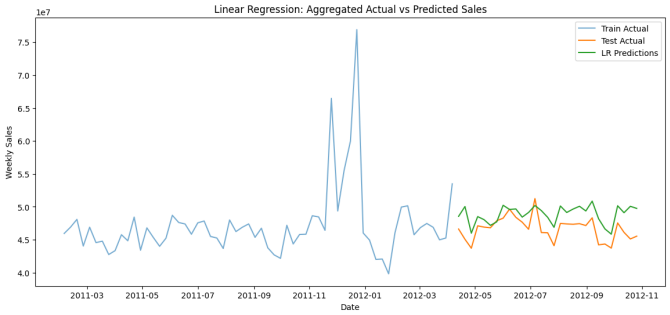


Fig. 10. Linear Regression predictions compared to actual values

Figure 10 presents the Linear Regression results. The model fails to capture both seasonal structure and variability, resulting in consistently poorer performance.

3) Random Forest Feature Importance:

TABLE VIII
TOP RANDOM FOREST FEATURE IMPORTANCES

Feature	Importance
lag_52	0.822
lag_1	0.135
rolling_mean_4	0.029
All others	< 0.020

This result (Table VIII) also explains why Random Forest performs similarly to Seasonal Naive and SARIMA: all three successful models rely, directly or indirectly, on the annual seasonal signal.

Figure 11 presents feature importance scores from the Random Forest model.

The results show a strong concentration of importance in lag-based features. The dominance of the 52-week lag confirms that the model captures annual seasonality through feature engineering.

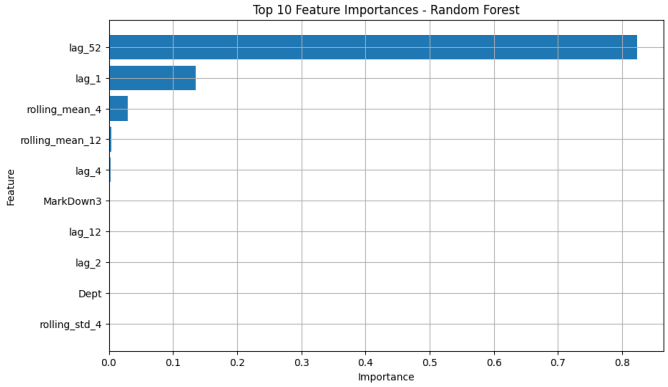


Fig. 11. Random Forest feature importance

This finding highlights that machine learning models rely heavily on engineered temporal features to replicate the seasonal structure that SARIMA models explicitly.

D. Consolidated Model Comparison

Table IX presents a summary of aggregated results for all models.

TABLE IX
FULL MODEL COMPARISON (AGGREGATED)

Model	MAE	RMSE	MAPE
SARIMA	1,034,871	1,443,201	2.19%
Random Forest	1,028,661	1,352,851	2.22%
Seasonal Naive	1,247,205	1,562,916	2.65%
Moving Average	1,591,870	1,901,638	3.44%
Naive	1,648,526	2,419,673	3.52%
ARIMA	1,673,011	2,136,956	3.62%
Linear Regression	2,263,158	2,576,671	4.91%

The results reveal a clear performance hierarchy. SARIMA and Random Forest form the top tier, with nearly identical performance. The Seasonal Naive model forms a strong second tier, confirming that capturing annual seasonality alone yields competitive forecasts.

The remaining models perform substantially worse, with ARIMA and Linear Regression consistently underperforming.

These results demonstrate that alignment between model structure and data characteristics, particularly seasonality, is the key determinant of forecasting accuracy, rather than model complexity alone.

IX. DISCUSSION

This section interprets the results presented in Section VIII in relation to the research questions defined in Section II. Rather than restating numerical findings, the discussion focuses on explaining the observed patterns and their methodological and practical implications.

A. Model Comparison: Answering the Main Research Question

The central research question examined whether classical time-series models or machine learning models provide more

accurate forecasts in a retail setting. The results do not support a definitive preference for either approach. Instead, they show that the best-performing classical model (SARIMA, MAPE = 2.19%) and the best-performing machine learning model (Random Forest, MAPE = 2.22%) achieve comparable predictive accuracy at the aggregated level. The difference between the two models is extremely small and not practically significant, and neither can be considered clearly superior under this experimental setup.

This finding challenges the assumption that machine learning methods automatically outperform classical statistical models in every forecasting context. As shown by [2], model performance depends heavily on data characteristics, particularly the amount of data and the strength of structural patterns. In this study, strong and stable seasonality allows a classical model such as SARIMA to perform as well as a more complex machine learning model.

At the same time, the similarity in performance should not be interpreted as equivalence in all contexts. The two models rely on fundamentally different mechanisms: SARIMA explicitly models seasonal structure, while Random Forest captures it implicitly through engineered features. Their relative performance may therefore differ under alternative conditions, such as longer time horizons or richer feature sets.

It is also important to note that the comparison between model families is not fully symmetric. Classical models are trained on aggregated data, while machine learning models are trained on disaggregated store–department data and subsequently aggregated. This difference in data structure and information availability should be considered when interpreting the results.

B. The Role of Seasonality

The results provide clear evidence that seasonality is the dominant factor driving forecasting accuracy in this dataset.

This is most clearly demonstrated by the contrast between ARIMA and SARIMA. The two models differ only in the inclusion of seasonal components, yet their performance differs substantially. ARIMA performs worse than all baseline models, while SARIMA achieves the lowest MAPE. This isolates seasonality as the key driver of improved accuracy.

A second line of evidence comes from the Seasonal Naive model. Despite having no parameters, it achieves strong performance by simply repeating values from the same week in the previous year. This highlights that capturing annual cycles alone provides a powerful predictive signal.

Together, these findings confirm that when demand exhibits strong and stable seasonal patterns, models that explicitly or implicitly capture seasonality will dominate those that do not.

C. The Role of Feature Engineering

The feature importance results from the Random Forest model provide a direct answer to the second research question.

The dominance of the 52-week lag feature indicates that the model relies primarily on past seasonal patterns to generate forecasts. This mirrors the behavior of SARIMA, which captures the same structure through its seasonal components.

This suggests that feature engineering can effectively substitute for explicit time-series modeling in settings where seasonal patterns are strong and stable. However, this substitution relies on the availability of well-designed lag features and may not generalize to more complex demand structures.

External variables appear to contribute relatively little compared with lag-based features. Despite their theoretical relevance, variables such as CPI, unemployment, and temperature contribute little to predictive performance in this dataset. This is consistent with the weak correlations observed in the EDA and likely reflects the limited time span and relatively stable macroeconomic conditions in the data.

D. Aggregated vs Disaggregated Evaluation

The results at the disaggregated level highlight an important methodological issue related to evaluation metrics.

At this level, MAPE values become extremely large and unreliable due to the presence of near-zero sales values. This is not indicative of model failure, but rather a known limitation of the metric itself.

Aggregation resolves this issue by producing a smoother series with strictly positive values, allowing MAPE to behave as intended. The dramatic difference between disaggregated and aggregated MAPE values illustrates that evaluation outcomes depend strongly on both the level of aggregation and the choice of metric.

This has practical implications. For applications focused on total demand forecasting, aggregated evaluation provides a more reliable basis for model comparison. For more granular forecasting tasks, alternative metrics may be required.

E. Business and Practical Interpretation

From a practical perspective, the results suggest that model selection in retail forecasting should not be based solely on accuracy.

SARIMA offers strong interpretability and performs well in settings with clear seasonal structure, but it is limited to univariate modeling and does not scale easily across many series.

Random Forest, on the other hand, can handle large panel datasets and incorporate multiple features, making it more flexible and scalable. However, its performance depends heavily on feature engineering, particularly the inclusion of lag variables.

Given the comparable accuracy of the two approaches, the choice between them should be guided by operational considerations such as scalability, interpretability, and data availability.

Overall, the results reinforce a key principle in forecasting: model effectiveness depends less on complexity and more on alignment with the underlying structure of the data.

X. Limitations

While this study provides a controlled comparison of forecasting approaches in a retail context, several limitations should be acknowledged.

A. Dataset Scope and Temporal Coverage

The dataset spans approximately three years, resulting in relatively short time series for each store–department combination. As shown by [2], the relative performance of machine learning models improves with larger datasets, while shorter series tend to favor classical statistical approaches.

As a result, the comparable performance of SARIMA and Random Forest observed in this study may partially reflect this data constraint rather than a general equivalence between the two approaches.

B. Lack of Hyperparameter Optimization

The Random Forest model was implemented using manually selected hyperparameters without systematic tuning. While this configuration provides reasonable performance, more extensive optimization could potentially improve results.

Similarly, other advanced machine learning models such as XGBoost or LightGBM were not fully explored under optimized conditions, and their performance may therefore be understated.

C. Modeling Asymmetry Across Approaches

The comparison between classical and machine learning models is not fully symmetric. Classical models were applied to aggregated data, while machine learning models were trained at the disaggregated store–department level and then aggregated.

Although this reflects practical constraints and typical usage patterns, it means that the two model families operate under different information structures, which should be considered when interpreting the results.

D. Evaluation Metric Limitations

MAPE is used as the primary metric for cross-model comparison due to its interpretability. However, it is known to be unstable in the presence of near-zero values and may produce distorted results at the disaggregated level.

Although this issue is mitigated by evaluating MAPE primarily at the aggregated level, reliance on a single metric may limit the robustness of conclusions. Future work could incorporate alternative metrics such as Mean Absolute Scaled Error (MASE).

E. Experimental Setting

The evaluation is conducted in a controlled, offline setting using historical data. While the chronological split ensures realistic forecasting conditions, it does not capture the full complexity of real-world deployment, including concept drift, data quality issues, and operational constraints.

As a result, the reported performance should be interpreted as indicative rather than definitive of real-world outcomes.

XI. Conclusion

This study evaluated and compared forecasting approaches across three methodological families: baseline methods, classical time-series models, and machine learning models, using a large-scale panel dataset of weekly Walmart retail sales. The primary objective was to determine which approach provides the most accurate forecasts under consistent experimental conditions and to identify the factors that drive differences in performance.

The results show that no single model family clearly dominates across all evaluation criteria. At the aggregated level, SARIMA (MAPE = 2.19%) and Random Forest (MAPE = 2.22%) achieve nearly identical predictive accuracy, indicating that classical and machine learning approaches can perform comparably when appropriately specified. This finding challenges the common assumption that machine learning methods systematically outperform statistical models and instead highlights the importance of aligning model structure with the underlying characteristics of the data.

Across all models, a clear pattern emerges: those that effectively capture annual seasonality consistently outperform those that do not. SARIMA achieves this through explicit seasonal modeling, while Random Forest captures the same structure through lag-based feature engineering. The strong performance of the Seasonal Naive model further reinforces that seasonality is the dominant driver of demand in this dataset. In contrast, ARIMA and Linear Regression fail to deliver competitive results, demonstrating that model sophistication alone does not guarantee improved forecasting accuracy.

The feature importance analysis confirms that machine learning models rely heavily on engineered temporal features, particularly the 52-week lag, to capture recurring patterns. This suggests that, in environments with stable seasonal structure, feature engineering can enable machine learning models to approximate the behavior of classical time-series approaches. However, the effectiveness of this strategy is likely to depend on the availability of sufficient historical data and well-designed features.

The study also highlights the importance of evaluation design. At the disaggregated level, MAPE becomes unstable due to near-zero values, producing misleading results. Aggregated evaluation provides a more reliable basis for model comparison, demonstrating that conclusions about model performance depend not only on the models themselves but also on how performance is measured.

Overall, the findings suggest that forecasting accuracy in retail demand is driven less by model complexity and more by the extent to which a model captures the dominant structural properties of the data. In this case, seasonality is the key factor, and both classical and machine learning approaches can perform effectively when this structure is properly modeled.

For practitioners, this implies that model selection should consider not only predictive accuracy but also factors such as interpretability, scalability, and data availability. A well-specified SARIMA model offers a simple and interpretable

solution for aggregate forecasting, while Random Forest provides a flexible and scalable alternative for panel data settings.

In conclusion, effective demand forecasting does not require the most complex models, but rather models that are well aligned with the data.

A. Future Work

While this study provides a controlled comparison of forecasting approaches, several directions for future research emerge.

First, extending the analysis to longer time horizons would provide a more comprehensive evaluation of model performance. As highlighted in this study, the relatively short time span of the dataset may favor classical time-series models. Evaluating models on longer and more recent datasets would help determine whether machine learning methods gain a stronger advantage as more data becomes available.

Second, incorporating exogenous variables into classical time-series models represents a natural extension. Models such as ARIMAX or SARIMAX could provide a more direct comparison with machine learning approaches by allowing external factors, including promotions and macroeconomic indicators, to influence forecasts.

Third, further improvements in machine learning performance could be achieved through systematic hyperparameter optimization and the exploration of more advanced models. While this study focused on Random Forest as a representative approach, additional experimentation with alternative algorithms and tuning strategies could yield further insights.

Fourth, future work could explore probabilistic forecasting methods that provide prediction intervals rather than point estimates. Such approaches are particularly relevant in retail contexts, where uncertainty estimation plays a critical role in inventory management decisions.

Finally, evaluating models in a real-world deployment setting would provide valuable validation of these findings. While this study uses a controlled offline evaluation, practical forecasting systems must account for challenges such as concept drift, real-time data availability, and operational constraints.

REFERENCES

[1] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*. San Francisco, CA, USA: Holden-Day, 1970.

[2] V. Cerqueira, L. Torgo, and C. Soares, "A case study comparing machine learning with statistical methods for time series forecasting: Size matters," J. Intell. Inf. Syst., vol. 59, pp. 415–433, 2022, doi: 10.1007/s10844-022-00713-9.

[3] M. S. Haque, M. S. Amin, and J. Miah, "Retail demand forecasting: A comparative study for multivariate time series," J. Math. Stat. Stud., vol. 4, no. 4, pp. 40–46, 2023, doi: 10.32996/jmss.2023.4.4.4.

[4] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

[5] IHL Group, "Retail inventory crisis persists despite $172 billion in improvements." [Online]. Available: https://www.ihlservices.com/news/analyst-corner/2025/09/retail-inventory-crisis-persists-despite-172-billion-in-improvements/

[6] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and machine learning forecasting methods: Concerns and ways forward," PLoS ONE, vol. 13, no. 3, Art. no. e0194889, 2018, doi: 10.1371/journal.pone.0194889.

[7] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M5 accuracy competition: Results, findings, and conclusions," Int. J. Forecast., vol. 38, no. 4, pp. 1346–1364, 2022, doi: 10.1016/j.ijforecast.2021.11.013.